Research Proposal: Computational Investigation of Spiral Volute Manifolds for Throttling Pintle Injectors

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1. Project Background & Motivation

In variable-thrust liquid rocket engines, movable pintle injectors often suffer from severe flow maldistribution during deep throttling. When propellant enters a standard annular manifold from a single side-port inlet, it creates torsional flow separation and unequal exit velocities, leading to localized thermal stresses and degraded combustion efficiency.

This study investigates whether adapting a spiral volute manifold—a geometry traditionally used in Francis turbines—can enforce uniform circumferential flow distribution and eliminate torsional swirl in micro-scale injector applications.

2. Phase 1: Completed Analytical Work (1D Sizing)

The foundational thermodynamics and 1D mathematical sizing for the manifold have been completed using NASA-GLENN CEA and custom MATLAB scripts. The analytical baseline is fully established:

- **Thermodynamic Design Point:** Extracted from NASA CEA for a sub-scale LOX/LCH4 engine running fuel-rich (O/F = 3.2, Pc = 20 bar, Thrust = 750 N).
- **Flow & Velocity Modeling:** A linear mass bleed-off model was formulated to size the cross-sectional area of the spiral channel at 1-degree increments, maintaining a strict constant-velocity target to prevent pressure imbalances.
- **Mathematical Boundary Constraints:** A 5% minimum-area cutoff at the 360-degree mark was implemented to prevent theoretical zero-thickness geometry, ensuring downstream mesh viability.
- **Viscous & Pressure Estimations:** Local Reynolds numbers were calculated, indicating highly turbulent flow (Re > 36,000). The Darcy-Weisbach equation with Blasius correlations predicts a total manifold pressure drop of roughly 0.4 bar, well within the standard 20% pressure-drop budget.

3. Current Challenges & Phase 2 Translation

With the theoretical sizing complete, the next phase requires constructing a 3D CAD model and running steady-state CFD simulations to test the 1D mathematical assumptions against 3D fluid behavior. Currently, two significant technical bottlenecks exist in transitioning from Phase 1 to Phase 2:

- **CAD Modeling Ambiguity:** Translating the MATLAB geometric arrays into a functional SolidWorks model has proven challenging due to an absence of properly labeled, comprehensive visual references for standard pintle injector internal geometry.
- **Foundational CFD Execution:** Establishing the complete CFD workflow—from defining the fluid domain and generating a viable mesh to configuring the solver—is a new frontier for this research.

4. Request for Mentorship & Guidance

Mentorship is sought to navigate the transition from mathematical theory to computational validation. The researcher aims to independently execute the modeling and simulation but would highly value high-level direction in the following core areas:

1. **Reference Geometry Definition:** Recommendations on sourcing valid CAD references or standard engineering drawings to accurately model the physical pintle injector components and manifold casing.
2. **Foundational CFD Methodology:** Methodological advice on establishing a standard CFD workflow from the ground up, including industry best practices for boundary condition application, mesh generation, and solver configuration.
3. **Data Validation:** Strategic direction on correctly extracting pressure and velocity outputs from the 3D CFD results to correlate them directly against the 1D MATLAB analytical predictions.

5. Expected Deliverables

The primary outcome of this research will be a comprehensive technical manuscript prepared for submission to a peer-reviewed academic publication. Recognizing the iterative nature of computational research, the final scope of the paper will remain flexible and adapt to the CFD findings. Potential research trajectories include:

- **A Comparative Validation Study:** Correlating the baseline 1D mathematical predictions against 3D CFD results to evaluate pressure drop and velocity uniformity.
- **A Design Optimization Analysis:** Exploring the parametric impact of internal geometry—such as varying the number of turning vanes or altering the minimum-area cutoff—on mitigating torsional flow.